Regd. Office **:**Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

NCERT BOOSTER TEST SERIES

Test – 07

Topics covered in various subjects:

Physics : Current Electricity, Moving charges and Magnetism, Magnetism & Matter

Chemistry : Surface Chemistry, General Principles and Processes of Isolation of Elements

Botany : Microbes in Human Welfare, Organisms and Populations, Ecosystem

Zoology : Biotechnology: Principles and Processes, Biotechnology and Its Applications

PHYSICS

Choose the correct answer:

1. In a closed circuit to maintain a steady current, an external agency moves electrical charge from
   (1) Lower to higher potential energy
   (2) Higher to lower potential energy
   (3) Lower to higher potential
   (4) Higher to lower potential
2. Electromotive force is
   (1) A force
   (2) The voltage difference between the two terminals of a source
   (3) The voltage difference between the two terminals of a source in closed circuit
   (4) The voltage difference between the two terminals of a source in open circuit
3. In ionic crystals and electrolyte liquids, the carrier of current may be
   (1) Electrons only
   (2) Negative ions only
   (3) Positive ions only
   (4) Positive and negative ions
4. Read the statements carefully and choose the correct option

   Statement A: Kirchhoff's junction rule is based on conservation of charge.

   Statement B: Bending or reorienting the wire changes the validity of Kirchhoff's junction rule.

   (1) Statement A is correct only
   (2) Statement B is correct only
   (3) Both the statements A and B are correct
   (4) Neither statement A nor B is correct
5. The storage battery of a car have emf of 16 V and internal resistance of 0.5Ω. If the terminals of the battery are short circuited, then the current from the battery is
   (1) 32 A (2) 16 A

(3) 8 A (4) Zero

6. A steady current flows in a metallic conductor of non-uniform cross-section. Which of these quantities is constant along the conductor?

(1) Current density (2) Drift speed

(3) Electric field (4) Current

7. The relation between V and I is not unique. It means there is more than one value of V for the same current I. A material exhibiting such behaviour is

(1) Copper (2) Aluminium

(3) Gallium arsenide (4) Diode

8. If a battery of emf 12 V and internal resistance 0.2Ω draws a maximum current, then the terminal voltage of the battery is

(1) 6 V (2) Zero

(3) 4 V (4) 12 V

9. A negligibly small current is passed through a wire of length 12 m and uniform cross-section 4.0×10^{-7} m^2, and its resistance is measured to be 4Ω. The resistivity of material at the temperature of the experiment is

(1) 4×10^{-7} Ωm (2) 3×10^{-7} Ωm

(3) 1.3×10^{-7} Ωm (4) 1.2×10^{-7} Ωm

10. The number density of free electrons in a copper conductor is 8.5×10^{28} m^{-3}. If area of cross-section of the wire is 10^{-6} m^2 and it is carrying a current of 1.0 A, then the time taken by the electrons to drift from one end of the wire 5.0 m long to its other end is

(1) 6.8×10^4 s (2) 6.8×10^3 s

(3) 1.68×10^4 s (4) 1.68×10^5 s

11. A battery of emf 12 V and internal resistance 2Ω is connected to a resistor. If the current in the circuit is 0.6 A, then terminal voltage of the battery is

(1) 12 V (2) 11 V

(3) 10.8 V (4) 13.2 V

12. A carbon resistor of (74 ± 7.4) kΩ is to be marked with rings of different colours for its identification.

The colour code sequence will be

(1) Yellow-green-violet-gold

(2) Green-orange-violet-gold

(3) Yellow-violet-orange-silver

(4) Violet-yellow-orange-silver

13. A storage battery of emf 12 V and internal resistance of 0.6 Ω is being charged by 124 V dc supply using a series resistor of 11.4 Ω. The terminal voltage of the battery is

(1) 17.6 V (2) 6.4 V

(3) 12 V (4) 124 V

14. The resistivity of a typical insulator amber is greater than that of a metal by a factor of the order of

(1) 10^{22} (2) 10^{11}

(3) 10^{19} (4) 10^{23}

15. On increasing the temperature, the resistivity of the alloy manganin

(1) Increases linearly

(2) Decreases

(3) Is nearly independent of temperature

(4) Increase exponentially

16. If n resistor each of resistance R are combined, then the ratio of the minimum to maximum effective resistance is

(1) n (2) $\frac{1}{n}$

(3) n^2 (4) $\frac{1}{n^2}$

17. In a potentiometer arrangement, a cell of emf 2.25 V gives a balance point at 45.0 cm length of the wire. If the cell is replaced by another cell and the balance point shifts to 99.0 cm. Then the emf of the second cell is

(1) 2.25 V (2) 4.95 V

(3) 4.25 V (4) 2.34 V

18. Six lead-acid type of secondary cells each of emf 3.0 V and internal resistance 0.02Ω joined in series to provide a supply to a resistance of 8.88Ω. The terminal voltage of the supply is

(1) 17.76 V (2) 11.92 V

(3) 16.35 V (4) 14.24 V

19. An unknown resistance R is connected to left gap of a meter bridge and another unknown resistance S connected to right gap, balance point from left end is at 30 cm. If now a resistance of 2Ω is connected in parallel with S, the null point occurs at 40 cm. The value of S is

(1) 2 Ω (2) 3.2 Ω

(3) 1.1 Ω (4) 4 Ω

20. A wire has a resistance of 1Ω at 27°C, and a resistance of 1.6Ω at 100°C. The temperature coefficient of resistance of the wire is nearly

(1) 0.008/°C (2) 0.009/°C

(3) 0.006/°C (4) 0.004/°C

21. The order of a magnitude of magnetic field on the earth's surface is

(1) 10^{-2} T (2) 10^{-1} T

(3) 10^{-5} T (4) 10° T

22. An electron (mass 9×10^{-31} kg and charge 1.6×10^{-19} C) is moving at a speed of 2×10^{6} m/s in a magnetic field of 2×10^{-4} T perpendicular to it. The radius of the path of the electron is

(1) 5.625 m (2) 56.25 m

(3) 562.5 m (4) 5.625 cm

23. Read the statements carefully and choose the correct option

Statement A: Electrostatic field lines originate at a positive charge and terminate at a negative charge or at infinity.

Statement B: Magnetic field lines always form closed loop.

(1) Only statement A is correct
(2) Only statement B is correct
(3) Both the statements A and B are correct
(4) Neither statement A nor B is correct

24. The smallest value of magnetic moment is

(1) 18.54×10^{-24} J/T (2) 9.27×10^{-14} J/T
(3) 9.27×10^{-24} J/T (4) 18.54×10^{-20} J/T

25. Read the statements carefully and choose the correct option

Statement A: Ampere's circuital law is independent of the Biot-Savart law.

Statement B: Ampere's circuital law can be derived from the Biot-Savart law.

(1) Only statement A is correct
(2) Only statement B is correct
(3) Both the statements A and B are correct
(4) Neither statement A nor B is correct

26. A solenoid of length 0.4 m has a radius of 1 cm and is made up of 400 turns. It carries a current of 4A. The magnitude of the magnetic field inside the solenoid is

(1) 5.2×10^{-3} T (2) 5.2×10^{-4} T
(3) 5.02×10^{-5} T (4) 5.02×10^{-3} T

27. The horizontal component of the earth's magnetic field at a certain place is 2×10^{-5} T and the direction of the field is from the geographic south to geographic north. If a very long straight conductor carrying steady current of 2A (from east to west) is placed on a horizontal table, then the force per unit length on it is

(1) 4×10^{-5} N upwards
(2) 4×10^{-5} N downwards
(3) 2×10^{-5} Nm^{-1} downwards
(4) 4×10^{-5} Nm^{-1} downwards

28. A circular coil of magnetic moment of 20A m^2 is placed in a vertical plane and is free to rotate about a horizontal axis which coincides with its diameter. A uniform magnetic field of 1T in the horizontal direction exists such that initially the axis of the coil is in the direction of the field. If the coil has rotated by 90°, then angular speed acquired by the coil is

(The moment of inertia of the coil is 0.1 kg m^2)

(1) 10 rad/s (2) 5 rad/s
(3) 20 rad/s (4) 40 rad/s

29. A circular coil of 20 turns and area of 0.2 m^2 carrying a current of 4A is suspended vertically in a uniform horizontal magnetic field of magnitude 1.0 T. The field lines make an angle 60° with the plane of the coil. The magnitude of the counter torque to prevent the coil from turning is

(1) 8 Nm (2) 4 Nm

(3) $8\sqrt{3}$ Nm (4) 80 Nm

30. Read the statements carefully and choose the correct option

Statement A: A moving coil galvanometer can be converted into an ammeter by introducing a shunt resistance of the small value in series.

Statement B: A moving coil galvanometer can be converted into a voltmeter by introducing a resistance of a large value in series.

(1) Only statement A is correct
(2) Only statement B is correct
(3) Both the statements A and B are correct
(4) Neither statement A nor B is correct

31. A charged particle enters an environment of a strong and non-uniform magnetic field varying from point to point both in magnitude and direction, and comes out of it following a complicated trajectory. If it suffered no

collisions with the environment, then its final speed is

(1) Equal to initial speed

(2) Two times the initial speed

(3) Half of the initial speed

(4) Different from initial speed

32. An electron travelling east to west enters a chamber having a uniform electrostatic field in south to north direction. If electron is not deflecting from its straight-line path, then the direction of a uniform magnetic field B is

(1) Vertically upward

(2) Vertically downward

(3) Towards North

(4) Towards South

33. A 3.0 cm wire carrying a current of 10 A is placed inside a solenoid parallel to its axis. The magnetic field inside the solenoid is given to be 0.27 T. The magnetic force on the wire is

(1) 2.1×10^{-2} N (2) 8.1×10^{-2} N

(3) Zero (4) 4.2×10^{-2} N

34. Two long and parallel straight wires A and B carrying currents of 4A and 2A in the same direction are separated by a distance of 2 cm. The force on a 10 cm section of wire A is

(1) 8×10^{-5} N (2) 8×10^{-7} N

(3) 8×10^{-6} N (4) 8×10^{-8} N

35. A galvanometer coil has a resistance of 10 Ω and the meter shows full scale deflection for a current of 2 mA. The resistance needed to convert into voltmeter of range 0 to 9 V, is

(1) 4490 Ω connecting in series

(2) 4500 Ω connecting in series

(3) 4490 Ω connecting in parallel

(4) 4500 Ω connecting in parallel

36. A short bar magnet of magnetic moment M = 0.15 J/T is placed in a uniform magnetic field of 0.15 T. The potential energy of the magnet in unstable equilibrium is

(1) 2.25×10^{-4} J (2) -2.25×10^{-2} J

(3) 2.25×10^{-2} J (4) -2.25×10^{-4} J

37. A short bar magnet placed with its axis at 37° with a uniform external magnetic field of 0.5 T experiences a torque of magnitude equal to 1.5×10^{-2} J. The magnitude of magnetic moment of the magnet is

(1) 5×10^{-2} J/T (2) 3.75×10^{-2} J/T

(3) 6×10^{-2} J/T (4) 1.73×10^{-2} J/T

38. A magnetic needle free to rotate in a vertical plane parallel to the magnetic meridian has its north tip pointing down at 60° with the horizontal. If the horizontal component of the earth's magnetic field at the place is 0.20 G. Then the earth's magnetic field at the place is

(1) 0.25 G (2) 0.15 G

(3) 0.23 G (4) 0.40 G

39. A short bar magnet has a magnetic moment of 0.24 JT^{-1}. The difference in the magnitude of the magnetic field at a distance of 10 cm from the centre of the magnet, on the axis and the equatorial line (normal bisector) of the magnet is

(1) 0.24 T (2) 0.48 T

(3) 0.24 G (4) 0.48 G

40. A short bar magnet of magnetic moment 3.75×10^{-2} JT^{-1} is placed with its axis perpendicular to the earth's field direction. If at a distance of 5 cm from the centre of the magnet, the resultant field is inclined at 45° with earth's field on its normal bisector then the magnitude of the earth's field is

(1) 0.3 T (2) 0.6 T

(3) 0.3 G (4) 0.6 G

41. A solenoid has a core of material with relative permeability 200. The windings of the solenoid are insulated from the core and carry a current of 4A. If the number of turns is 500 per metre, then the magnetic intensity is

(1) 1 A/m (2) 2×10^3 A/m

(3) 0.5 A/m (4) 0.1 A/m

42. Read the statements carefully and choose the correct option

Statement A: Magnetic field arises due to charges in motion.

Statement B: A system can have magnetic moments even though its net charge is zero.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements A and B are correct

(4) Neither statement A nor B is correct

43. The geographic meridian is defined as

(1) The vertical plane containing the latitude circle and the axis of rotation of earth

(2) The horizontal plane containing the latitude circle and the axis of rotation of earth

(3) The vertical plane containing the longitude circle and the axis of rotation of earth

(4) The horizontal plane containing the longitude circle and the axis of rotation of earth

44. Read the statements carefully and choose the correct option

Statement A: Magnetostatic field lines can never form closed loops around empty space.

Statement B: A closed loop of static magnetic field line must enclose a region across which a current is passing.

(1) Only statement A is correct
(2) Only statement B is correct
(3) Both the statements A and B are correct
(4) Neither statement A nor B is correct

45. The magnetic susceptibility is positive for
(1) Diamagnetic material only
(2) Paramagnetic material only
(3) Ferromagnetic material only
(4) Paramagnetic and ferromagnetic material

CHEMISTRY

46. Consider the following statements about adsorption
(a) It causes decrease in residual forces on the surface
(b) It is an endothermic process
(c) It is accompanied by decrease in entropy of the system

The correct statements are
(1) (a) and (b) only (2) (b) and (c) only
(3) (a) and (c) only (4) (a), (b) and (c)

47. Incorrect statement about physisorption is
(1) It is reversible in nature
(2) It arises because of van der Waals' forces
(3) It is highly specific in nature
(4) It results into multimolecular layer

48. For Freundlich adsorption isotherm, slope of the graph, $\log \frac{x}{m}$ vs log P is
(1) log k (2) $\log\frac{1}{k}$
(3) n (4) $\frac{1}{n}$

49. Decomposition of $KClO_3$ is accelerated in presence of
(1) Silica gel (2) Fe
(3) Al_2O_3 (4) MnO_2

50. In the Haber's process for manufacture of ammonia, promoter used in the reaction is
(1) Molybdenum (2) Nickel
(3) Palladium (4) Copper

51. Hydrolysis of $C_{12}H_{22}O_{11}$ in presence of catalytic amount of H_2SO_4 gives
(1) Glucose and Mannose
(2) Glucose and Fructose
(3) Glucose only
(4) Fructose only

52. Consider the following reaction

$$\text{Vegetable oils} + A \xrightarrow{Ni} \text{Vegetable ghee}$$

A in the reaction is
(1) H_2 (2) H_2SO_4
(3) O_2 (4) NO

53. Product obtained when carbon monoxide reacts with hydrogen in presence of copper catalyst is
(1) Formaldehyde (2) Methyl alcohol
(3) Formic acid (4) Methane

54. Consider the following statements about zeolites
(a) They have honeycomb like structure
(b) They are used as catalysts in petrochemical industries
(c) ZSM-5 is used to convert gasoline into alcohols

The incorrect statement(s) is/are
(1) (a) and (b) only (2) (a) and (c) only
(3) (c) only (4) (a) only

55. Diastase converts starch into
(1) Ethyl alcohol (2) Lactose
(3) Maltose (4) Glucose

56. In contact process for the manufacture of sulphuric acid the catalyst used is
(1) V_2O_5 (2) Fe_2O_3
(3) $AlCl_3$ (4) Cu_2Cl_2

57. Which among the following is an example of gel?

(1) Cell fluids (2) Butter

(3) Cheese (4) Froth

58. Consider the following statements

(a) Sol of gelatine is a reversible sol

(b) Gold sol is lyophilic sol

(c) Lyophobic sols need stabilising agents for their preservation

The correct statements are

(1) (a) and (b) only (2) (b) and (c) only

(3) (a) and (c) (4) (a), (b) and (c)

59. Which among the following is multimolecular colloid?

(1) Sulphur sol

(2) Sol of starch

(3) Sol of proteins

(4) Sol of sodium stearate

60. The given reaction for the preparation of As_2S_3 sol is called

$As_2O_3 + 3H_2S \rightarrow As_2S_3(sol) + 3H_2O$

(1) Hydrolysis

(2) Oxidation

(3) Reduction

(4) Double decomposition

61. Which among the following is/are method(s) of preparation of colloidal sol?

(1) Electrophoresis (2) Peptization

(3) Dialysis (4) Both (1) and (2)

62. For hair cream, dispersed phase and dispersion medium respectively are

(1) Liquid and solid (2) Solid and liquid

(3) Liquid and liquid (4) Solid and gas

63. For soaps, the critical micelle concentration range is

(1) 1.0 to 0.5 M (2) 10^{-4} to 10^{-3} M

(3) 10^{-6} to 10^{-4} M (4) 1 to 2 M

64. Positively charged sol among the following is

(1) Methylene blue sol (2) Silver sol

(3) CdS sol (4) Congo red sol

65. Consider the following statements

(a) When KI solution is added to $AgNO_3$ solution, negatively charged sol is obtained

(b) Combination of two layers of opposite charges around the colloidal particle is called Helmholtz electrical double layer

(c) Colligative properties shown by colloidal particles are of small order as compared to values shown by true solutions of same concentrations

The correct statement(s) is/are

(1) (a) and (b) only (2) (b) and (c) only

(3) (a) and (c) only (4) (c) only

66. In coagulating a positive sol, the order of flocculating power of $PO_4^{3-}, NO_3^-, SO_4^{2-}$ and $[Fe(CN)_6]^{4-}$ is

(1) $NO_3^- > SO_4^{2-} > PO_4^{3-} > [Fe(CN)_6]^{4-}$

(2) $PO_4^{3-} > [Fe(CN)_6]^{4-} > SO_4^{2-} > NO_3^-$

(3) $SO_4^{2-} > PO_4^{3-} > NO_3^- > [Fe(CN)_6]^{4-}$

(4) $[Fe(CN)_6]^{4-} > PO_4^{3-} > SO_4^{2-} > NO_3^-$

67. In coagulating gold sol, the coagulating power of which ion is highest?

(1) Al^{3+} (2) Ba^{2+}

(3) PO_4^{3-} (4) Br^-

68. Which among the following is an ore of iron?

(1) Siderite (2) Malachite

(3) Calamine (4) Bauxite

69. Which among the following is not used as collector in froth floatation process?

(1) Cresols (2) Pine oils

(3) Fatty acids (4) Xanthates

70. Reagent used in leaching alumina from bauxite is

(1) NaOH (2) H_2SO_4

(3) NaCN (4) NH_4OH

71. The metal which is used for galvanising iron is

(1) Zinc (2) Copper

(3) Nickel (4) Cobalt

72. Metal which is refined by Mond's process is

(1) Ni (2) In

(3) Al (4) Zr

73. In van Arkel method for refining metal, titanium is heated with

(1) CO (2) I_2

(3) Br_2 (4) Charcoal

74. In Brass, percentage of Zn and Cu respectively are

(1) 60% and 40% (2) 75% and 25%

(3) 40% and 60% (4) 25% and 75%

75. In extraction of gold the reducing agent used is

(1) Coke (2) Zn

(3) Fe (4) CO

76. Which among the following is an example of homogeneous catalysis?

(1) $N_2(g) + 3H_2(g) \xrightarrow{Fe(s)} 2NH_3(g)$

(2) $4NH_3(g) + 5O_2(g) \xrightarrow{Pt(s)} 4NO(g) + 6H_2O(g)$

(3) $CO(g) + 3H_2(g) \xrightarrow{Ni(s)} CH_4(g) + H_2O(g)$

(4) $CH_3COOCH_3(l) \xrightarrow{HCl(l)} CH_3COOH(aq) + CH_3OH(aq)$

77. The metal which is not purified by zone refining is

(1) Gallium (2) Indium

(3) Germanium (4) Aluminium

78. Copper matte contains

(1) Cu_2O and FeS (2) Cu_2S and FeS

(3) Cu_2S and FeO (4) Cu and Cu_2S

79. Carbon content in cast iron is approximately

(1) 4% (2) 3%

(3) 6% (4) 1%

80. In the extraction of iron using Blast furnace, which of the given reactions does not occur at (500-800) K temperature range?

(1) $FeO + CO \rightarrow Fe + CO_2$

(2) $Fe_2O_3 + CO \rightarrow 2FeO + CO_2$

(3) $3Fe_2O_3 + CO \rightarrow 2Fe_3O_4 + CO_2$

(4) $Fe_3O_4 + 4CO \rightarrow 3Fe + 4CO_2$

81. In extraction of iron using blast furnance flux used is

(1) Limestone (2) Silica

(3) Alumina (4) Haematite

82. Which metal is refined by distillation?

(1) Mercury (2) Tin

(3) Chromium (4) Lead

83. Metal which is absent in german silver is

(1) Cr (2) Cu

(3) Zn (4) Ni

84. Metal which is extracted using self reduction method is

(1) Al (2) Fe

(3) Cu (4) Zn

85. Blister copper is obtained due to evolution of

(1) CO_2 (2) SO_2

(3) SO_3 (4) H_2S

86. Consider the following statements about froth floatation method

(a) This method is employed for removing gangue from sulphide ores

(b) Aniline is used as froth stabiliser

(c) Mineral particles become wet by water while gangue particles by oils

The correct statements are

(1) (a) and (b) only (2) (b) and (c) only

(3) (a) and (c) only (4) (a), (b) and (c)

87. Depressant used in the separation of ZnS from PbS is

(1) NaCN (2) NaOH

(3) Phenol (4) Ammonia

88. Which one of the given methods is not employed in concentration of ores?

(1) Hydraulic washing

(2) Magnetic separation

(3) Leaching

(4) Roasting

89. Water vapours are adsorbed by

(1) Anhydrous calcium chloride

(2) NaCl

(3) Charcoal

(4) Silica gel

90. The metal which is refined by liquation is

(1) Tin (2) Silver

(3) Gold (4) Zinc

BOTANY

91. All of the given statements are **true** w.r.t. lactic acid bacteria (LAB), **except**

(1) Grow in milk and convert it to curd

(2) Produce acid that coagulate and partially digest the milk proteins

(3) Improves nutritional quality of curd by increasing Vitamin E content

(4) Check disease - causing microbes in our stomach

92. The puffed-up appearance of dough during making foods such as dosa and idli is due to

(1) Multiplication of bacteria

(2) Production of CO_2 gas

(3) Death of yeast

(4) Addition of black gram

93. Which of the given food/beverage products is obtained with the help of bacteria?

(1) Swiss cheese (2) Roquefort cheese

(3) Bread (4) Beer

94. Which of the given alcoholic drinks are obtained without distillation of fermented broth?

(1) Whisky (2) Brandy

(3) Rum (4) Wine

95. Read the following statements and select the option which is true for them.

Statement A: Antibiotics are chemical substances, which are produced by some microbes and can kill or retard the growth of other (disease-causing microbes).

Statement B: Antibiotic means 'against life' (in the context of disease-causing organisms); whereas with reference to human beings, they are 'pro-life' and not against.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements A and B are correct

(4) Both the statements A and B are incorrect

96. Match the **Column-I** with **Column-II** and select the **correct** option

	Column-I	**Column-II**
A.	*Aspergillus niger*	(i) Ethanol
B.	*Acetobacter aceti*	(ii) Butyric acid
C.	*Clostridium butylicum*	(iii) Citric acid
D.	*Saccharomyces cerevisiae*	(iv) Acetic acid

	A	**B**	**C**	**D**
(1)	(iii)	(iv)	(ii)	(i)
(2)	(i)	(ii)	(iii)	(iv)
(3)	(iv)	(iii)	(ii)	(i)
(4)	(ii)	(i)	(iv)	(iii)

97. Which of the given bioactive molecules is/are produced by bacteria?

(A) Streptokinase, (B) Cyclosporin-A, (C) Statins

(1) Only (A) (2) Both (A) and (C)

(3) Both (B) and (C) (4) Only (C)

98. Which of the given bioactive molecules has been commercialised as blood-cholesterol lowering agent?

(1) Penicillin (2) Amylase

(3) Statin (4) Protease

99. Match the **Column-I** with **Column-II** w.r.t. bioactive molecules and their uses and select the **correct** option

	Column-I	**Column-II**
A.	Lipase	(i) Clot buster
B.	Pectinase	(ii) Detergent formulation
C.	Streptokinase	(iii) Immunosuppressive agent
D.	Cyclosporin A	(iv) Clarifying bottled juices

	A	**B**	**C**	**D**
(1)	(iv)	(iii)	(ii)	(i)

(2)	(i)	(ii)	(iii)	(iv)
(3)	(iii)	(i)	(iv)	(ii)
(4)	(ii)	(iv)	(i)	(iii)

100. Read the statements (A-C) stating them true (T) or false (F) and select the **correct** option.

A A major component of waste water is human excreta.

B The municipal waste-water is also called sewage.

C Sewage contains large amount of organic matter and microbes.

(1) A-T, B-T, C-T (2) A-F, B-T, C-F

(3) A-T, B-T, C-F (4) A-F, B-F, C-T

101. During sewage treatment, which of the given microorganisms are involved in flocs formation?

(1) Aerobic bacteria and fungi

(2) Anaerobic bacteria and fungi

(3) Symbiotic bacteria and algae

(4) Unicellular algae and anaerobic fungi

102. During primary treatment of sewage, all solids that settle forms the A , and the supernatant forms the B .

Select the **correct** option to fill 'A' and 'B'.

(1) A – Primary sludge, B – Primary effluent

(2) A – Primary effluent, B – Primary sludge

(3) A – Activated sludge, B – Flocs

(4) A – Flocs, B – Activated sludge

103. Select the **incorrect** statement w.r.t. 'Baculoviruses'.

(1) Are pathogen that attack insects and other arthropods.

(2) Majority of Baculoviruses used as biological control agents are in the genus *Nucleopolyhedrovirus*.

(3) Are excellent candidates for species-specific, broad spectrum insecticidal application.

(4) Have no negative impacts on plants, mammals, birds, fish or even on non-target insects.

104. During sewage treatment, which of the given bacteria is commonly found in anaerobic sludge?

(1) *Anabaena*

(2) *Methanobacterium*

(3) *Nostoc*

(4) *Oscillatoria*

105. Which of the given gases is predominantly found in biogas?

(1) CH_4 (2) CO_2

(3) H_2 (4) H_2S

106. The study of the interactions among organisms and between the organism and its physical (abiotic) environment is called

(1) Genetics (2) Ecology

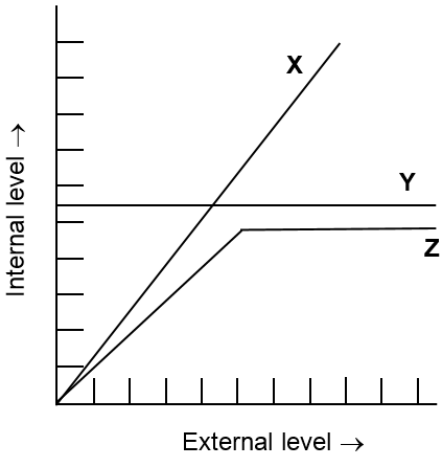
(3) Morphology (4) Physiology

107. The salinity (in parts per thousand) is less than 5 in

(1) Inland water (2) Sea

(3) Hypersaline lagoons (4) Ocean

108. Refer the figure given below showing organismic response to abiotic factors. Identify X, Y and Z respectively and select the **correct** option



	X	Y	Z
(1)	Regulators	Partial regulators	Conformers
(2)	Partial regulators	Conformers	Regulators
(3)	Conformers	Regulators	Partial regulators
(4)	Regulators	Conformers	Partial regulators

109. In aquatic animals, the osmotic concentration of body fluids change with that of ambient water osmotic concentration. To which of the given group, these organisms belong?
(1) Regulator (2) Conformer
(3) Partial regulator (4) Eurythermal

110. Mammals from colder climates generally have shorter ears and limbs to minimise heat loss. This is called the
(1) Allen's rule (2) Jordan's rule
(3) Rensch's rule (4) Bergmann's rule

111. Consider the following four conditions (a-d) and select the **correct** pair of them as adaptation to environment in Kangaroo rat.
(a) Meets all its water requirement through its internal fat oxidation
(b) Basks in sun when temperature is low
(c) Ability to concentrate its urine
(d) Presence of thick layer of fat below the skin
(1) (a) and (b) (2) (b) and (c)
(3) (c) and (d) (4) (a) and (c)

112. If in a pond there are 50 fishes last year and through reproduction 20 new fishes are added, taking the current population to 70, calculate the birth rate (in terms of offsprings per fishes per year).
(1) 0.4 (2) 0.1
(3) 0.2 (4) 0.5

113. Number of death in the population during a given period is called
(1) Natality (2) Mortality
(3) Immigration (4) Emigration

114. Which type of growth status of human population is reflected by the age pyramid shown below in the figure?
(1) Stable population
(2) Declining population
(3) Vanishing population
(4) Expanding population

115. Which of the given pairs w.r.t. population attribute contribute to an increase in population density?
(1) Mortality and Natality
(2) Mortality and Emigration
(3) Natality and Emigration
(4) Natality and Immigration

116. The population grows in an exponential or geometric fashion
(1) When resources in the habitat are unlimited
(2) When there is a competition between individuals for limited resource
(3) Results in a sigmoid growth curve
(4) When the population density reaches the carrying capacity

117. The integral form of exponential population growth is expressed by the equation
(1) $N_t = N_0 e^{rt}$ (2) $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$
(3) $\frac{dN}{dt} = rN\left(\frac{1-N}{K}\right)$ (4) $\frac{dN}{dt} = rN$

118. If '+' sign is assigned to beneficial interaction, '–' sign for detrimental and '0' for neutral interaction, then the population interaction represented by '+' '0' refers to
(1) Mutualism (2) Amensalism
(3) Competition (4) Commensalism

119. Match the Column-I with Column-II w.r.t. population interactions

Column-I	Column-II
A. Predation	(i) Fig tree and wasp
B. Parasitism	(ii) Visiting flamingoes and resident fishes in South American lakes
C. Mutualism	(iii) *Cuscuta* growing on hedge plants

D. Competition (iv) Tiger and deer

	A	B	C	D
(1)	(iv)	(iii)	(i)	(ii)
(2)	(i)	(ii)	(iii)	(iv)
(3)	(iii)	(i)	(iv)	(ii)
(4)	(ii)	(iv)	(i)	(iii)

120. Some species have evolved adaptation to avoid unfavourable condition by escaping in time includes all of the given, **except**

(1) Migration (2) Aestivation

(3) Hibernation (4) Diapause

121. Which of the given is an aquatic ecosystem?

(1) Forest (2) Grassland

(3) Desert (4) Estuary

122. Vertical distribution of different species occupying different levels in an ecosystem is called

(1) Stratification (2) Homeostasis

(3) Food chain (4) Food web

123. Net primary productivity is the available biomass for consumption to which of the given trophic levels and/or organisms?

(A) T_1 (B) T_2

(C) T_3 (D) Decomposers

(1) Only (A) (2) Both (B) and (D)

(3) Only (D) (4) Both (B) and (C)

124. The rate of formation of new organic matter by consumers is called

(1) Gross primary productivity

(2) Primary productivity

(3) Net primary productivity

(4) Secondary productivity

125. The annual net primary productivity of the whole biosphere is approximately A (dry weight) of organic matter. Select the **correct** option which fills 'A'.

(1) 170 billion tonnes (2) 55 billion tonnes

(3) 115 billion tonnes (4) 155 billion tonnes

126. Select the **incorrect** statement w.r.t. 'Humus'.

(1) It is a dark coloured amorphous substance

(2) It is highly resistant to microbial action

(3) It undergoes decomposition at an extremely fast rate

(4) It is colloidal in nature

127. Which of the given factors favours the decomposition of detritus?

(A) Sugar and nitrogen rich detritus

(B) Warm and moist environment

(C) Anaerobiosis

(D) Low temperature

(1) (A) and (B) (2) (C) and (D)

(3) (B) and (C) (4) (A) and (D)

128. The base of each ecological pyramid represents the A while the apex represents B .

Select the option which **correctly** fills 'A' and 'B'.

(1) A – Top level consumer, B – Producer

(2) A – Producer, B – Top level consumer

(3) A – Primary consumer, B – Producer

(4) A – Tertiary consumer, B – First trophic level

129. Read the following statements and select the option which is **true** for them.

Statement A: Pyramid of energy is always upright.

Statement B: The pyramid of biomass in sea is generally inverted.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements A and B are correct

(4) Both the statements A and B are incorrect

130. Which of the given organisms is primary as well as secondary consumer?

(1) Grasshopper (2) Cow

(3) Goat (4) Sparrow

131. Xerarch succession progress from

(1) Xeric to mesic condition

(2) Mesic to xeric condition

(3) Mesic to hydric condition

(4) Hydric to mesic condition

132. During succession, pioneer communities on rocks are usually

(1) Lichens (2) Bryophytes

(3) Shrubs (4) Forest

133. What percentage of PAR is captured by plants?

(1) 100% (2) 50%

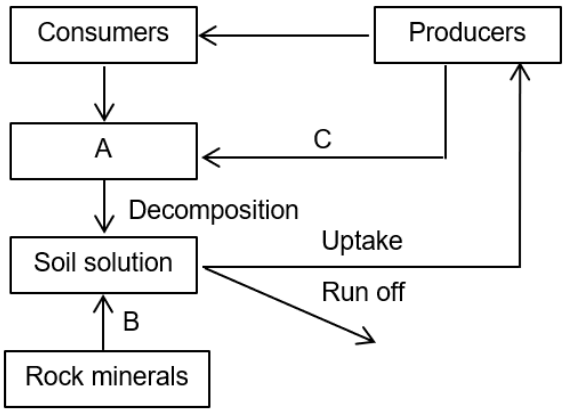
(3) 2-10% (4) 1-5%

134. In which of the given areas, secondary succession begins?

(1) Burned forest (2) Bare rock

(3) Newly created pond (4) Newly cooled lava

135. Refer the given simplified model of phosphorus cycling in a terrestrial ecosystem. Identify A, B and C respectively and select the **correct** option



	A	B	C
(1)	Weathering	Detritus	Litter fall
(2)	Litter fall	Detritus	Weathering
(3)	Detritus	Weathering	Litter fall
(4)	Decomposers	Litter fall	Detritus

ZOOLOGY

136. Read the following statements carefully.

(a) Stanley Cohen, a Stanford scientist, studied some useful restriction enzymes of *E.coli* bacterium

(b) Herbert Boyer, at the University of California at San Francisco, developed a method of removing plasmids and reinserting in other cells

(c) Biotechnological products brought qualitative improvement in health and food production

(d) First recombinant DNA was made by Stanley Cohen and Herbert Boyer in the year 1972

How many of the above statements are **correct**?

(1) Four (2) Three

(3) Two (4) One

137. All of the following techniques are included in biotechnology **except**

(1) Developing a DNA vaccine

(2) Curd making

(3) *In vitro* fertilisation leading to a 'test-tube baby

(4) Improving crop varieties by classical plant breeding

138. EFB definition of biotechnology encompasses

(1) Genetic engineering only

(2) Traditional breeding only

(3) Modern molecular biotechnology only

(4) Both traditional and modern molecular biotechnology

139. Read the following statements and choose the option which **correctly** represents True(T) and False(F) statements.

(a) Traditional hybridisation procedures often lead to inclusion of undesirable genes

(b) Biotechnology allows inclusion of only desirable genes

(c) Bioprocess engineering involves maintenance of microbial contamination free ambience

(d) An alien piece of DNA is linked with origin of replication site so that it can make multiple identical copies

	(a)	(b)	(c)	(d)
(1)	T	T	T	F
(2)	T	T	T	T
(3)	T	F	T	F
(4)	F	F	T	T

140. Choose the **mismatch.**

(1) Basmati rice – 27 varieties in India
(2) Rosie cow – 1997
(3) Patent on Basmati rice by an American company – 1997
(4) First clinical gene therapy – 1901

141. The linking of antibiotic resistance gene with the plasmid vector became possible with the enzyme
(1) Restriction endonuclease
(2) DNA ligase
(3) DNA polymerase
(4) Reverse transcriptase

142. First recombinant DNA was made by linking ___I___ gene with plasmid of ____II____. Choose the correct option for I and II.

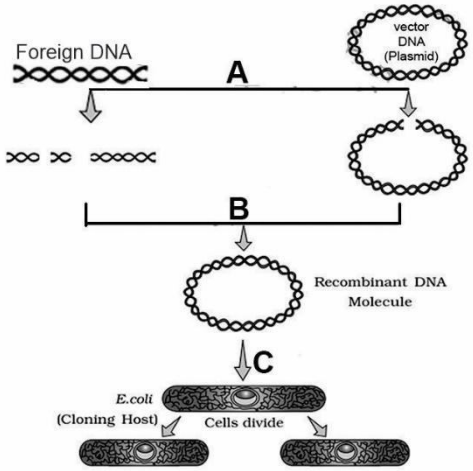
	I	II
(1)	Antibiotic sensitive	*Salmonella typhimurium*
(2)	Antibiotic resistant	*Escherichia coli*
(3)	Antibiotic resistant	*Salmonella typhimurium*
(4)	Antibiotic sensitive	*Escherichia coli*

143. Enzymes responsible for restricting the growth of bacteriophage in *E. coli* were isolated in the year
(1) 1963 (2) 1972
(3) 1990 (4) 1997

144. Choose the **odd** one w.r.t. *Hin*d II.
(1) First restriction endonuclease
(2) Recognition sequence is 8 bp long
(3) Isolated and characterised five years later after the discovery of restriction enzymes
(4) Isolated from *Haemophilus* influenzae

145. Which of the following **correctly** represents the site at which *Eco*R I makes cut?
(1) 5'— GAA↓TTC— 3'
3'— CTT↑AAG— 5'
(2) 5'— G↓AATTC— 3'
3'— C↑TTAAG— 5'
(3) 5'— G↓AATTC— 3'
3'__ CTTAA↑G__ 5'
(4) 5'— GAA↓TTC— 3'
3'— CTTAA↑G— 5'

146. Given below is a diagrammatic representation of recombinant DNA technology. Choose the option which **correctly** represents **A**, **B** and **C** respectively

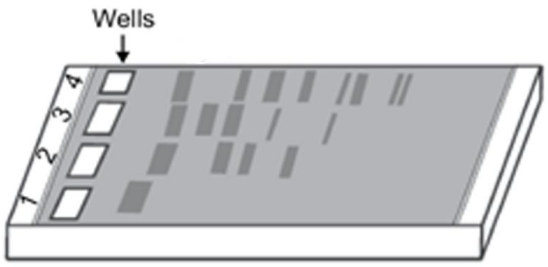


	A	B	C
(1)	Same restriction endonuclease	DNA ligases	Cloning
(2)	Different restriction endonuclease	DNA polymerase	Cloning
(3)	Same restriction exonuclease	*Taq* polymerase	Transformation
(4)	Same restriction endonuclease	DNA ligases	Transformation

147. If a restriction enzyme cuts the strand of DNA a little away from the centre of the palindromic site, then which type of ends will **not** produce?
(1) Sticky ends
(2) Single stranded portions at the ends
(3) Overhanging stretches on each strand
(4) Blunt ends

148. DNA fragments are _____charged molecules, and move towards _____ under an electric field. Choose the option which **correctly** fill the blanks respectively.
(1) Negatively, cathode
(2) Positively, anode
(3) Negatively, anode
(4) Positively, cathode

149. A typical agarose gel electrophoresis is shown below, identify the lane in which smallest fragment is present and choose the **correct** option



(1) Lane 1 (2) Lane 2
(3) Lane 3 (4) Lane 4

150. Match the Column-I with Column-II and choose the **correct** option

	Column-I		Column-II
(a)	Extraction of DNA from the gel piece	(i)	Seaweeds
(b)	Separation of DNA fragments	(ii)	Elution
(c)	Staining of DNA	(iii)	Agarose gel electrophoresis
(d)	Agarose gel	(iv)	Ethidium bromide

(1) a-iii, b-ii, c-iv, d-i
(2) a-ii, b-iii, c-iv, d-i
(3) a-iv, b-iii, c-i, d-ii
(4) a-i, b-ii, c-iii, d-iv

151. Read the following statements and choose the **correct** option
Statement A: Copy number of bacteriophages is very high.
Statement B: Copy number depends on *ori* gene.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both the statements A and B are correct
(4) Both the statements A and B are incorrect

152. If we ligate alien piece of DNA in a plasmid whose copy number is 15, then how many copies of alien piece of DNA we can obtain per cell?
(1) 15 (2) 30
(3) 150 (4) 60

153. Select the **correct** statement.
(1) Native *E.coli* cells are resistant to chloramphenicol and Kanamycin
(2) A cloning vector should have many recognition sites for single restriction enzyme
(3) Presence of cloning sites for different restriction enzymes in a cloning vector is a desirable feature
(4) Recognition site of *Pvu* II is present in amp^R in pBR322

154. If we introduce alien piece of DNA at *Hin*d III restriction site in cloning vector pBR322, then the resultant recombinants will show
(a) Resistance to ampicillin
(b) Resistance to tetracycline
(c) Sensitivity to ampicillin
(d) Sensitivity to tetracycline
Which of the above are **correct**?
(1) a and b (2) c and d
(3) a and d (3) b and c

155. Select the **incorrect** statement w.r.t. cloning vectors.
(1) Replication starts from *ori* sequence
(2) *Rop* gene codes for the proteins involved in the replication of the plasmid
(3) The ligation of alien DNA is done at a restriction site which is not present in the selectable marker gene
(4) Size of vector should be small

156. A research student was using pBR322 as cloning vector. He ligated gene of interest at *Sal* I site. Based on this, choose the **correct** option.

(1) Non-transformants will grow on both ampicillin and tetracycline containing medium
(2) Transformants will grow on ampicillin containing medium
(3) Non-recombinants will not grow on tetracycline containing medium
(4) Recombinants will not grow on ampicillin containing medium

157. One antibiotic resistance gene helps in selecting the __**A**__, whereas the other antibiotic resistance gene gets inactivated due to insertion of alien DNA, helps in selection of __**B**__. Choose the option which **correctly** fill the blanks.

	A	**B**
(1)	Transformants	Recombinants
(2)	Recombinants	Transformants
(3)	Non-recombinants	Non-transformants
(4)	Non-transformants	Non-recombinants

158. In blue-white screening of recombinants, gene of interest is ligated in the coding sequence of
(1) Ampicillin (2) Tetracycline
(3) β-galactosidase (4) Kanamycin

159. *Agrobacterium tumifaciens*, a pathogen of several __**A**__ plants is able to deliver a piece of DNA known as __**B**__. Choose the option which **correctly** fill the blanks.

	A	**B**
(1)	Dicot	Ti plasmid
(2)	Monocot	Ti plasmid
(3)	Dicot	T-DNA
(4)	Monocot	T-DNA

160. Which of the following vector is suitable for delivering a DNA fragment into animal cells?
(1) *A. tumifaciens*
(2) Retrovirus
(3) pBR322
(4) Bacteriophage

161. Select the **correct** sequence of making bacterial cells competent.
(a) Incubation of bacterial cells with recombinant DNA on ice
(b) Treatment with Ca^{+2} ions
(c) Heat shock at 42°C
(d) Putting them back on ice
(1) a → c → d → b
(2) b → a → c → d
(3) b → d → a → c
(4) a → d → b → c

162. All of the following are methods of direct gene transfer **except**
(1) Micro-injection
(2) Disarmed pathogen vectors
(3) Biolistic
(4) Gene gun

163. Read the following statements
(a) DNA is a hydrophobic molecule
(b) In micro-injection, DNA is injected into nucleus of an animal
(c) In Biolistics, cells are bombarded with high velocity micro-particles of DNA coated with gold or tungsten
(d) Gene gun is suitable for plant cells
How many of the above statements are **correct**?
(1) Three (2) Four
(3) Two (4) One

164. Arrange the following steps of process of recombinant DNA technology in **correct** sequence and choose the **correct** option.
(a) Isolation of desired DNA fragment
(b) Fragmentation of DNA by restriction endonucleases
(c) Formation of recombinant vector
(d) Large scale production of host cells
(e) Transferring the recombinant DNA into the host
(1) a → b → c → e → d
(2) b → a → c → e → d
(3) b → c → a → d → e
(4) a → b → c → d → e

165. Choose the **mismatch** w.r.t. isolation of DNA.
(1) Bacteria – Lysozyme
(2) Fungus – Chitinase
(3) Plant cells – Cellulase
(4) Animal cell – DNase

166. In PCR, __**A**__ bonds break during denaturation step, __**B**__ bonds are formed during annealing step and __**C**__ bonds are formed by *Taq* polymerase in extension step. Identify **A**, **B** and **C** and choose the correct option.

	A	**B**	**C**
(1)	Hydrogen bonds	Hydrogen bonds	Phosphodiester bonds
(2)	Phosphodiester	Hydrogen	Sugar-phosphate

	bonds	bonds sugar bonds	
(3)	Sugar-Hydrogen	Phosphodiester	
	phosphate sugar bonds	bonds bonds	
(4)	Phosphodiester Bonds	Hydrogen bonds bonds	Hydrogen

167. In which of the following culture system cells are maintained in their physiologically most active log/exponential phase?
(1) Closed culture system
(2) Continuous culture system
(3) Batch fermentation
(4) Culture system in which nutrients are added only once

168. Select the **incorrect** statement w.r.t. downstream processing.
(1) Large scale production of host cells
(2) Include separation and purification
(3) Quality control testing
(4) Done after biosynthetic stage

169. All of the following are applications of biotechnology **except**
(1) Recombinant therapeutics
(2) Bioremediation
(3) Tailor made plants
(4) Disease diagnosis by conventional methods

170. GM crops are useful in many ways **except**
(1) Reduced post harvest losses
(2) Reduced efficiency of mineral usage by plants which prevents early exhaustion of fertility of soil
(3) Reduced reliance on chemical pesticides
(4) Crops are made tolerant to abiotic stresses

171. Bt toxin is produced by
(1) *Thermus aquaticus*
(2) *Bacillus thuringiensis*
(3) *Agrobacterium tumifaciens*
(4) *Meloidegyne incognitia*

172. Bt toxin does not kill the bacteria which produces it because
(1) Bacteria are resistant to the toxin
(2) Toxin remains inactive in bacteria
(3) Toxin exist as nucleic acid crystal
(4) Toxin is enclosed in a protective sac

173. Select the **incorrect** w.r.t. Bt cotton.
(1) Alkaline pH of the cotton bollworm solubilize the crystals
(2) Activated toxin binds to the mouth parts and create pores in cotton bollworm
(3) Protein encoded by *cry I Ac and cry II Ab* control cotton bollworms
(4) Bt toxins are insect group specific

174. Read the following statements and choose the **correct** option
Statement A: RNA interference (RNAi) takes place in all prokaryotic organisms as a method of cellular defense.
Statement B: RNAi prevents transcription of the mRNA.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both the statements A and B are correct
(4) Both the statements A and B are incorrect

175. *Meloidogyne incognitia* could not survive in a transgenic tobacco expressing
(1) Specific sense mRNA only
(2) Specific interfering RNA
(3) Specific interfering protein
(4) Specific interfering nucleotide

176. Choose the **incorrect** statement w.r.t. SCID.
(1) It is a hereditary disease
(2) It is due to duplication of the gene for ADA
(3) Can be treated by bone marrow transplantation
(4) ADA enzyme is crucial for the immune system to function

177. Which host was used by Eli Lilly company to produce human insulin?
(1) *Entamoeba coli*
(2) *Escherichia coli*
(3) *Salmonella typhimurium*
(4) Transgenic mice

178. Which of the following is **not** correct w.r.t. the advantage of human insulin over insulin extracted from pancreas of slaughtered cattle and pigs?
(1) Do not induce unwanted immunological responses
(2) Free from risk of infection
(3) Large scale production is not possible

(4) Structure of recombinant human insulin is absolutely identical to that of natural molecule

179. Match Column-I with Column-II and choose the **correct** option

Column-I	**Column-II**
(a) α-1 antitrypsin	(i) Transgenic mice
(b) α-lactalbumin	(ii) Emphysema
(c) Safety of polio vaccine	(iii) Cystic fibrosis
(d) Transgenic model	(iv) Rosie cow

(1) a-ii, b-iv, c-i, d-iii
(2) a-iii, b-iv, c-ii, d-i
(3) a-i, b-ii, c-iii, d-iv
(4) a-iv, b-i, c-ii, d-iii

180. Match Column-I with Column-II and choose the **correct** option

Column-I	**Column-II**
(a) GEAC	(i) Early diagnosis of disease
(b) Biopiracy	(ii) Validity of GM research
(c) ELISA	(iii) Antigen-antibody interaction
(d) Recombinant DNA technology	(iv) Unlawful use of bio resources

(1) a-ii, b-iv, c-i, d-iii
(2) a-ii, b-iv, c-iii, d-i
(3) a-i, b-ii, c-iii, d-iv
(4) a-iv, b-ii, c-iii, d-i

❑❑❑